

Lecture #8

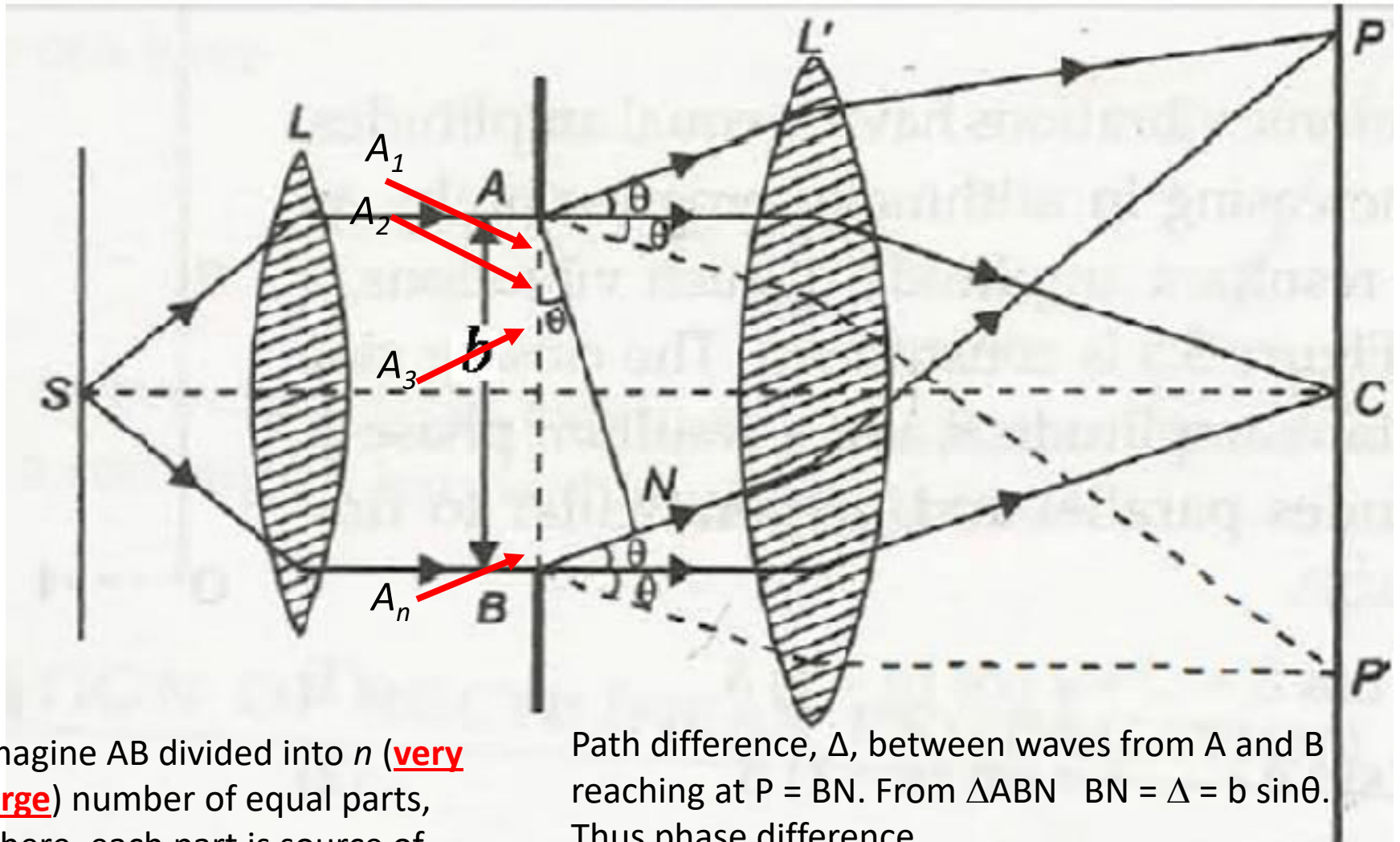
Content to be covered

- ❑ Single slit diffraction
 - ❖ Expression for resultant intensity
 - ❖ Position of maxima (Central and Secondary)
 - ❖ Position of minima
 - ❖ Intensity at secondary maxima
 - ❖ Spread of central diffraction maximum

Resources to be consulted

- ❑ OPTICS by Ajay Ghatak

Single Slit Diffraction



Imagine AB divided into n (**very large**) number of equal parts, where, each part is source of secondary wavelets.

Path difference, Δ , between waves from A and B reaching at P = BN. From $\triangle ABN$ $BN = \Delta = b \sin \theta$. Thus phase difference

$$\delta = \frac{2\pi}{\lambda} b \sin \theta$$

Single Slit Diffraction

The phase difference between any two successive parts of slit AB would be $\phi = \frac{1}{n} \left[\frac{2\pi}{\lambda} b \sin \theta \right]$

Thus if disturbance at point P due to point A_1 is $a \cos \omega t$ then from point A_2 : $a \cos (\omega t - \phi)$; from point A_3 : $a \cos (\omega t - 2\phi)$; from point A_N : $a \cos (\omega t - (N-1)\phi)$

Thus, resultant disturbance at point P would be

$$E = a \cos \omega t + a \cos(\omega t - \phi) + \dots + a \cos(\omega t - (n-1)\phi)$$

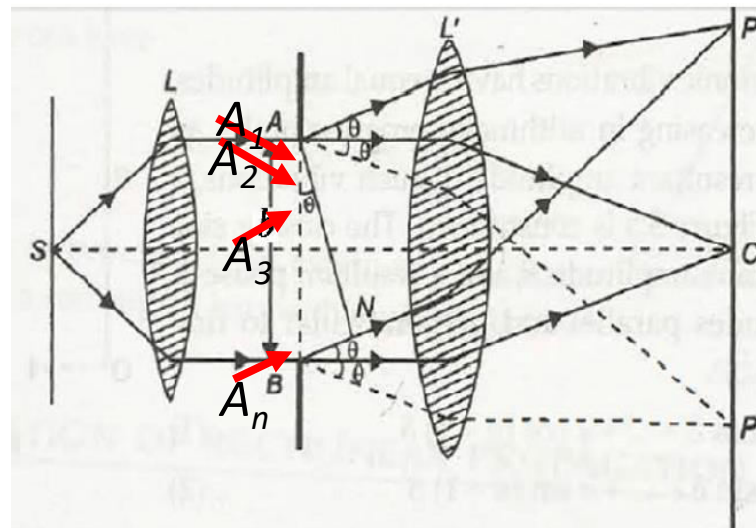
$$E = R \cos \left[\omega t - \frac{(n-1)\phi}{2} \right]$$

where $R = \frac{a \sin \frac{n\phi}{2}}{\sin \frac{\phi}{2}}$

$$R = \frac{a \sin \left(\frac{\pi b \sin \theta}{\lambda} \right)}{\sin \left(\frac{\pi b \sin \theta}{n\lambda} \right)}$$

Let $\beta = \pi b \sin \theta / \lambda$

$$R = \frac{a \sin \beta}{\sin (\beta/n)}$$



Single Slit Diffraction

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where

$$R = \frac{a \sin \frac{n\phi}{2}}{\sin \frac{\phi}{2}}$$

$$R = \frac{a \sin \left(\frac{\pi b \sin \theta}{\lambda} \right)}{\sin \left(\frac{\pi b \sin \theta}{n\lambda} \right)}$$

Let $\beta = \pi b \sin \theta / \lambda$

$$R = \frac{a \sin \beta}{\sin (\beta/n)}$$

But for value of n, $\sin \frac{\beta}{n} \approx \frac{\beta}{n}$

Therefore $R = \frac{a \sin \beta}{\beta/n}$

$$\Rightarrow R = \frac{A \sin \beta}{\beta} \quad \text{where } A = na$$

Thus resultant intensity at P

$$I = R^2 = A^2 \frac{\sin^2 \beta}{\beta^2} = I_0 \frac{\sin^2 \beta}{\beta^2}$$

Position Of Maxima And Minima In Single Slit Diffraction

Principal/Central Maxima

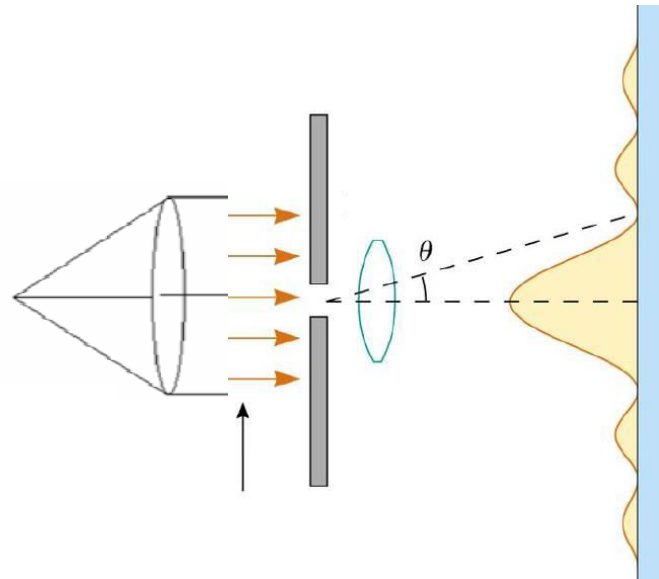
The resultant amplitude is given by $R = \frac{A \sin \beta}{\beta}$

Thus R will be maximum when $\beta = 0 \rightarrow \mathbf{R = A = na}$

$$I_{\max} = A^2 = I_0$$

$$\text{and } \beta = \frac{\pi b \sin \theta}{\lambda} = 0 \Rightarrow \theta = 0$$

This is the location of central maxima.



Position Of Maxima And Minima In Single Slit Diffraction

Minima

Re-consider the equation $R = \frac{A \sin \beta}{\beta}$

The resultant amplitude R will be minimum when $\sin \beta = 0$ but $\beta \neq 0$

i.e. $\beta = \pm\pi, \pm2\pi, \pm3\pi, \pm4\pi \dots, \pm n\pi$

$$\Rightarrow \beta = \pm n\pi \quad \text{where } n = 1, 2, 3, \dots$$

$$\text{Hence } \beta = \frac{\pi b \sin \theta}{\lambda} = \pm n\pi$$

$$\Rightarrow \boxed{b \sin \theta = \pm n\lambda}$$

where $n = 1, 2, 3, \dots$ gives the condition of 1st, 2nd, 3rd, minima. Here $n \neq 0$, because for $n = 0$, $\theta = 0$ which corresponds central maxima.

This is the condition for minima.

Position Of Maxima And Minima In Single Slit Diffraction

Let us consider the condition for minima $b \sin \theta = \pm n\lambda \Rightarrow \theta = \pm \sin^{-1}\left(\frac{n\lambda}{b}\right)$

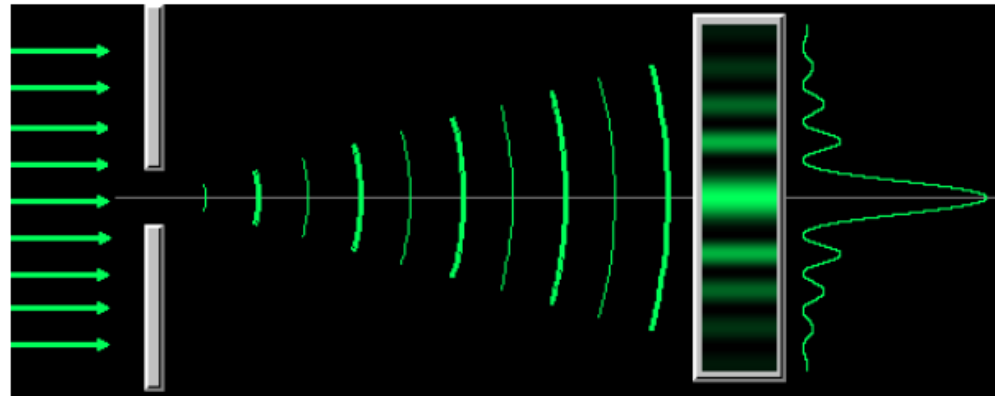
Position of first minima would be $\theta = \pm \sin^{-1}\left(\frac{\lambda}{b}\right)$

Position of second minima would be $\theta = \pm \sin^{-1}\left(\frac{2\lambda}{b}\right)$

Further $\sin \theta = \pm \frac{n\lambda}{b}$

since $\sin \theta \leq 1$

$$n \leq \frac{b}{\lambda}$$



Position Of Maxima And Minima In Single Slit Diffraction

Secondary Maxima

Apply the method of finding maxima and minima

$$I = R^2 = A^2 \frac{\sin^2 \beta}{\beta^2}$$

Differentiating above equation with respect to β

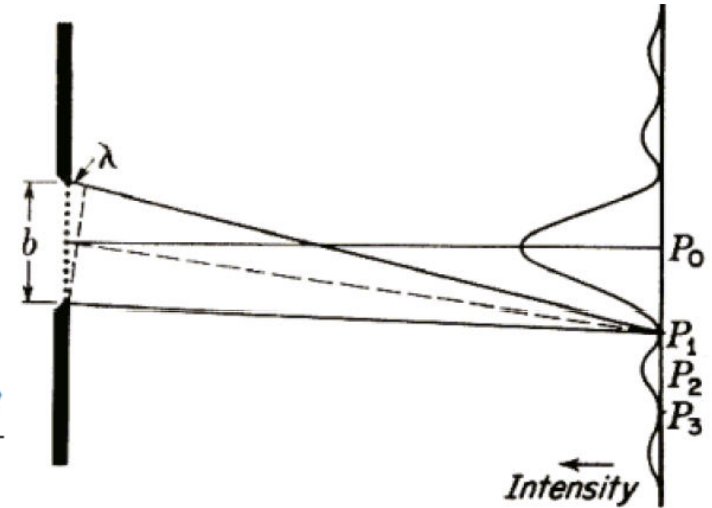
$$\therefore \frac{dI}{d\beta} = \frac{d}{d\beta} \left(\frac{A^2 \sin^2 \beta}{\beta^2} \right) = A^2 \frac{\beta^2 \times 2 \sin \beta \cos \beta - \sin^2 \beta \times 2\beta}{\beta^4}$$

$$\Rightarrow \frac{dI}{d\beta} = A^2 \left[\frac{2\beta \sin \beta \cos \beta - 2 \sin^2 \beta}{\beta^3} \right]$$

Making $\frac{dI}{d\beta} = 0$ We have, $A^2 \frac{2 \sin \beta}{\beta} \left[\frac{\beta \cos \beta - \sin \beta}{\beta^2} \right] = 0$

Thus either $\frac{\sin \beta}{\beta} = 0$ or $\frac{\beta \cos \beta - \sin \beta}{\beta^2} = 0$

$\Rightarrow \sin \beta = 0$ (minima) $\beta \cos \beta - \sin \beta = 0 \Rightarrow \beta = \tan \beta$



But here $\sin \beta = 0$ or $\beta = m\pi$ gives the position of primary m^{th} minima, except when $\beta = 0$.

Position Of Maxima And Minima In Single Slit Diffraction

Thus, the position of secondary maxima is given by the equation:

$$\beta = \tan \beta \quad (\text{secondary maxima})$$

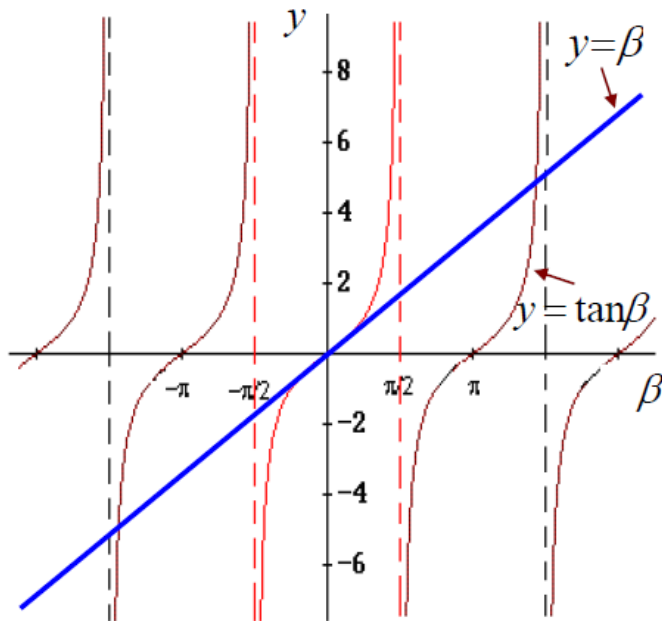
Here, $\beta=0$ corresponds to the central maximum.

Note: The other roots can be obtained by determining the points of intersections of the curves, $y = \beta$ and $y = \tan \beta$ by graphical method.

From the graph, other roots which satisfy the equation are

$$\beta = \pm(2n+1)\frac{\pi}{2} \quad \text{where } n = 1, 2, 3, \dots$$

The more exact values are 1.430π , 2.46π , 3.47π , 4.471π



Position Of Maxima And Minima In Single Slit Diffraction

Thus, directions of secondary maxima are approximately:

$$\beta = \pm(2n+1)\frac{\pi}{2}$$

$$\Rightarrow \frac{\pi b \sin \theta}{\lambda} = \pm(2n+1)\frac{\pi}{2}$$

$$\Rightarrow \theta = \pm \sin^{-1} \left[\frac{(2n+1)\lambda}{b} \right]$$

So, secondary maxima lie at $b \sin \theta = 3\lambda/2, 5\lambda/2, 7\lambda/2$ etc as first, second, third etc....

Intensity Calculations

Intensity of diffraction pattern is given as,

$$I = R^2 = A^2 \frac{\sin^2 \beta}{\beta^2}$$

Intensity of Principal maximum: $\beta = 0$

$$I_{\max} = A^2 = I_0$$

The intensity of secondary maxima

Substituting $\beta = \pm \frac{3\pi}{2}$ (first secondary maxima i.e. for $n = 1$)

$$I_1 = A^2 \frac{\sin^2 \frac{3\pi}{2}}{\left(\frac{3\pi}{2}\right)^2} = \frac{4A^2}{9\pi^2} \Rightarrow I_1 = \frac{I_0}{22} = 0.046I_0$$

Thus Intensity of first secondary maxima (I_1) is 4.6% of principal maxima (I_0).

Intensity Of Maxima And Minima In Single Slit Diffraction

Similarly, for intensity of second secondary maxima

let us substitute,

$$\beta = \pm \frac{5\pi}{2}$$
$$I_2 = A^2 \frac{\sin^2 \frac{5\pi}{2}}{\left(\frac{5\pi}{2}\right)^2} = \frac{4A^2}{25\pi^2} \Rightarrow I_2 = \frac{I_0}{62} = 0.016I_0$$

Thus Intensity of second secondary maxima (I_2) is 1.6% of principal maxima (I_0).

I_3 is 0.83% of I_0 .

One can similarly calculate $I_3, I_4, I_5, \dots$. From the above values of intensities, it is obvious that intensity of secondary maxima decreases as the order increases.

Ratio of Principal maximum to secondary maxima:

$$I_0 : I_1 : I_2 : I_3 : \dots = 1 : 4/9\pi^2 : 4/25\pi^2 : 4/49\pi^2 \dots$$

Spread of central diffraction maximum

Direction of first minima is given by

$$b \sin \theta = \pm \lambda, \quad \text{or} \quad \sin \theta = \pm \frac{\lambda}{b}$$

If the lens is very near to the slit, then $L \approx f$.

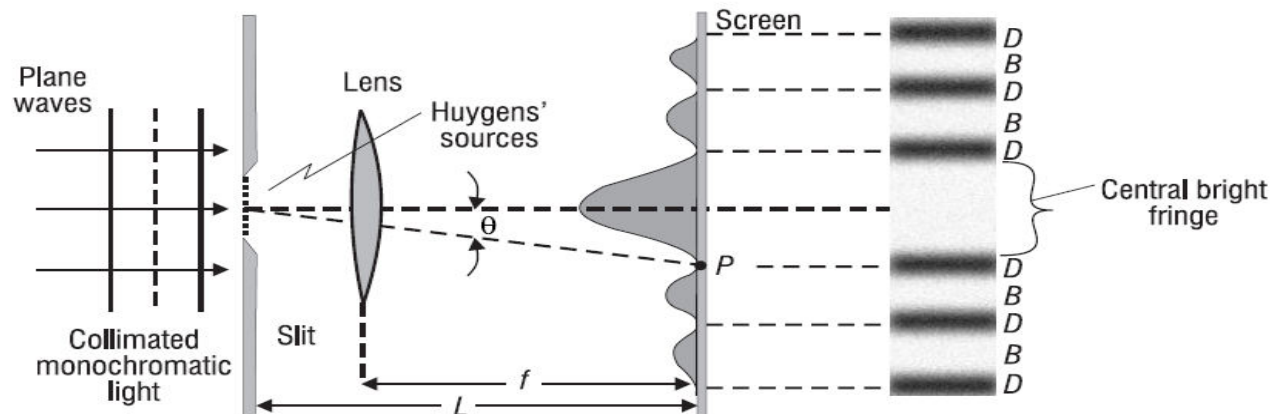
$$\tan \theta \sim \sin \theta = \frac{y}{L}$$

where, y is the linear half width of the central maxima and f is the focal length of the lens.

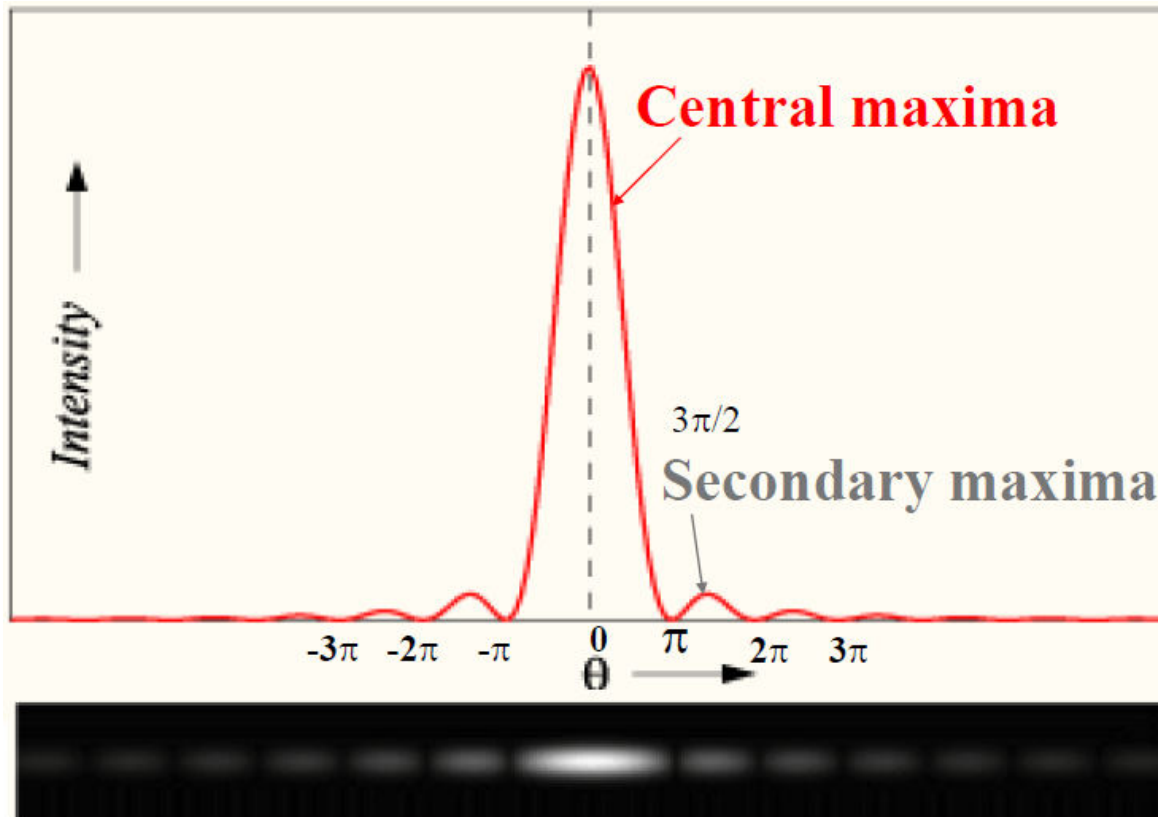
$$\frac{y}{f} = \frac{\lambda}{b}$$

Hence, the width of central Maxima is given by

$$2y = \frac{2f\lambda}{b}$$



Diffraction pattern from a single slit: Angular positions of maxima and minima



Single Slit Fraunhofer diffraction: Effect of slit width

Principle maxima, $\beta = 0$, $I = I_0 = A^2$

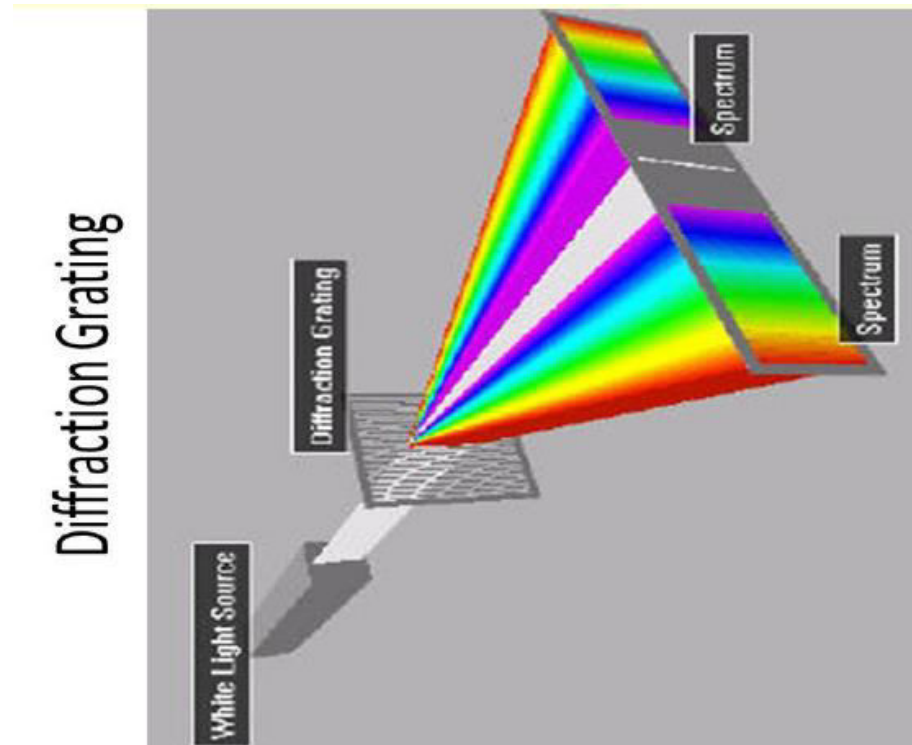
Minima, $\sin \beta = 0$, $\beta = n\pi$

$$\sin \theta = \pm \frac{n\lambda}{b}$$

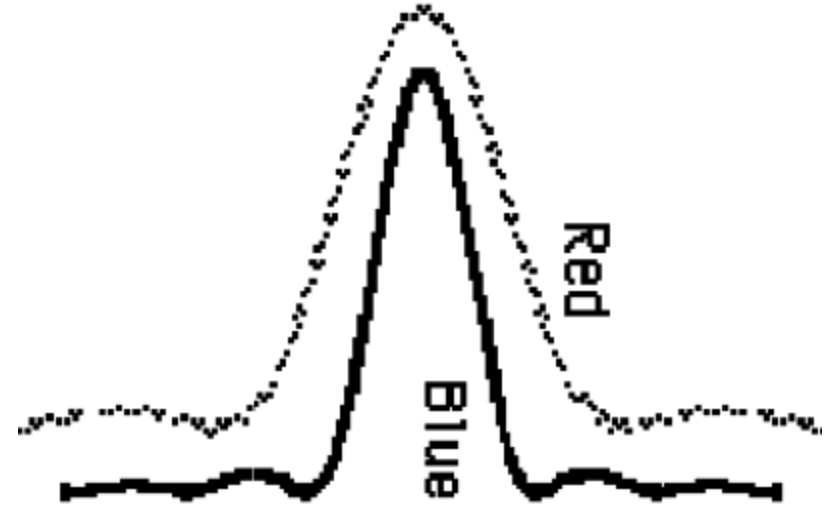
First minima at $\sin \theta = \pm \lambda/b$ and

$$\sin \theta \sim \frac{y}{D}$$

$$y_{\min.} = \frac{mD\lambda}{b}$$



Single Slit Diffraction



The longer wavelength is diffracted more than the shorter one.

Narrower slits diffract more than wider ones

Slits farther : Maxima closer

Slits closer : Maxima farther

Single Slit Diffraction vs Double slit Interference

